

Reflection on AI-Assisted Scientific Writing

Assignment 4 | Aman Kumar Emaley | MCL2026012

Converting the manuscript into a structured LaTeX document was easiest for the prose sections because the source paper already had a clear academic organization. The most manual effort was required for the benchmark table and the conceptual TikZ figure: long cell text had to be constrained to the A4 text width, while figure nodes and arrows had to remain readable without crossing the margins. The most useful Prism feature was task-specific code generation with explicit preservation instructions. Prompts that named the required environment, label, caption, and placement reduced mechanical editing and made the result easier to inspect. The AI-assisted output still required human checking. In particular, citation keys, bibliography completeness, table wording, and figure placement could not be accepted solely because the source compiled. I standardized the keys in `main.tex`, prepared a matching `references.bib` file from the supplied references, and kept the scientific claims and numerical content unchanged. I also documented likely undefined-reference and table-overflow problems and their targeted corrections. The provided project screenshots show the sequential workflow from manuscript conversion through table, figure, and Overleaf-oriented checks. The final package is designed for portability: it contains the source, bibliography, original paper, figure/evidence files, prompt log, error log, and reflection. Researchers should treat AI as a mechanical assistant rather than an author: compare every claim with the original paper, verify citations against the publication record, inspect equations and tables cell by cell, compile more than once, and retain responsibility for scientific accuracy and authorship.

Submission status: Complete package assembled with the requested source, PDF, bibliography, evidence screenshots, prompt log, error log, and reflection.